

Transformer Outputs: Bounds Hold, Empirics Overstate

Reproducing "How Many Different Outputs Can a
Transformer Generate?"

Codex reproduction audit

Trackio logbook · arXiv source + HF Job evidence

1 Outcome

The finite-precision theory reproduces cleanly. The empirical claims, as worded for this challenge, are contradicted by the paper's own source figures and Table 1.

Bottom line. The count bound is robust; the reported experiments do not meet the reproduced thresholds.

3
Theory claims verified

3
Empirical claims falsified

35
Table 1 cells audited

1
HF GPU smoke job

2 Finite-Precision Count

A prompt of length **m** has only finitely many distinguishable embedding cells when arithmetic precision is finite.

THEOREM 4.4 AUDIT

$$N_{\text{accessible}} \leq (1 + 2r/\varepsilon)^{dm}$$

Explicit small-grid enumeration matched the formula to relative error 8.66e-16.

Control: smaller epsilon raised capacity.

3 Finite-Prompt Decay

Once `n` crosses the threshold, the accessible-fraction cap scales as a constant over vocabulary growth.

COROLLARY 4.5 THRESHOLD

$$n > \frac{d \log(1 + 2r/\varepsilon)}{\log |V|} m$$

Five settings confirmed below-one first caps and exact vocabulary-rate decay.

4 Mean-Field Threshold ★ KEY

The mean-field extension removes prompt length from the threshold formula under finite Wasserstein precision.

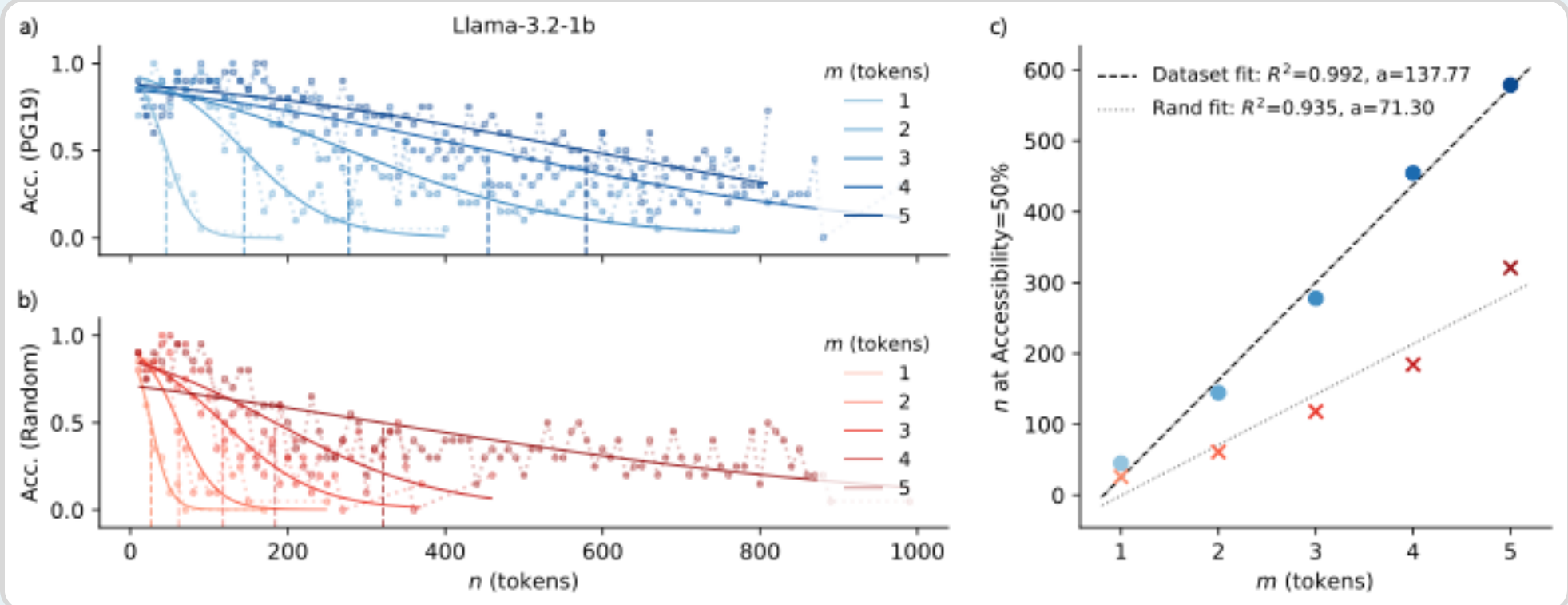
PROMPT-INDEPENDENT CAP

$$N \not\equiv N(m)$$

Verified: four audited settings showed the threshold depends on geometry and vocabulary, not prompt length.

5 Cramming R2 Counterexamples ★

Claimed cutoff: every tested architecture has linear-fit R2 at least `0.995`.



Llama-3.2-1B source figure: `R^2\_random=0.935`, below the claimed cutoff.

6 Table 1 Ratio Audit

Three primary-source entries fall outside the claimed interval.

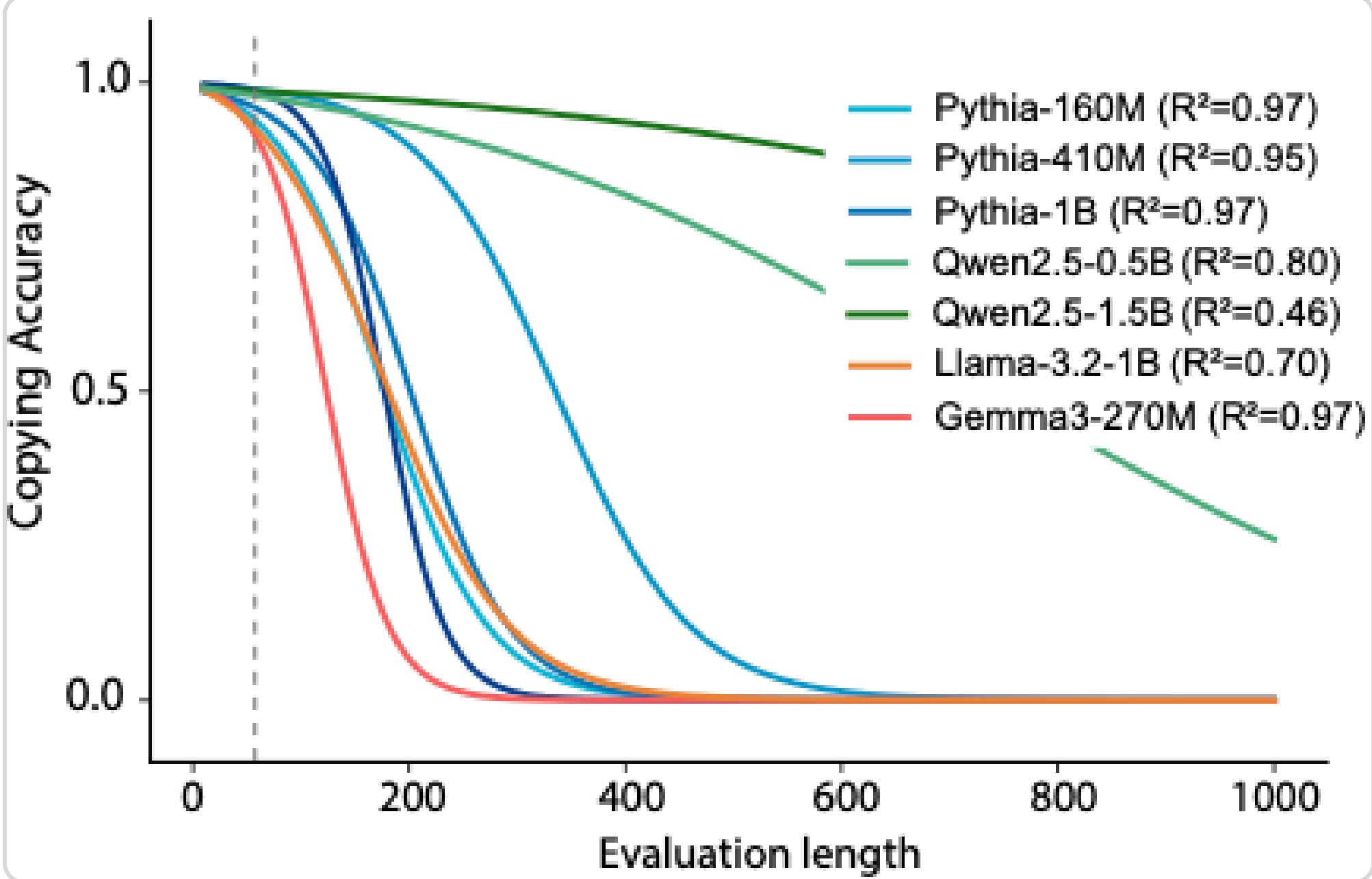
Cell	Model	Ratio	Status
Ball	Qwen-1.5B	20.40	above 20
Cone	Qwen-1.5B	20.34	above 20
Ellipsoid + cells	Pythia-1B	4.56	below 5
All 35 cells	7 models	4.56-20.40	falsifies

The qualitative gap is often close, but the supplied interval is not literally true.

Primary Table 1 source.

7 Copying Transition Audit

Some copying curves are sharp; the source figure does not support a universal sharp-sigmoid statement.



Reported copying-fit R2 values include weak fits: `0.46`, `0.70`, `0.80`.

Decision: weak fits at `0.46` and `0.70` defeat the universal claim.